BCA

FINANCIAL ACCOUNTING AND MANAGEMENT

UNIT-5

WORKING CAPITAL MANAGEMENT

Meaning of Working Capital

Working capital is the part of the company's total capital. This capital is required by all the businesses for financing their short-term needs and is a part of current assets. It is the measure of the company's efficiency to pay its short-term dues as well as manage operational expenses.

Mathematically,

Working Capital=Current Assets-Current Liabilities

Definitions of Working Capital, as per various management experts are as under:

1. "Working Capital is the excess of C.A. over current liabilities." - H.G, Guthmann

2. "Working Capital is descriptive of that capital which is not fixed. But the more common use of the Working Capital is to consider it as the difference between the book value of the C.A. and current liabilities." - Hoglend. J. Bierman, and A. K. Mc Adams,.

3. "Working Capital represents the excess of C.A. over current liabilities" - J.L. Brown and L.R. Housard.

Nature of Working Capital

Working capital management is concerned with the problems that arise in attempting to manage the current assets, the current liabilities and the interrelations that exist between them. Current assets refer to those assets which in the ordinary course of business can be, or will be, converted into cash within one year without undergoing a diminution in value and without disrupting the operations of the firm. Examples- cash, marketable securities, accounts receivable and inventory. Current liabilities are those liabilities which are intended, at the inception, to be paid in the ordinary course of business, within a year, out of the current assets or the earnings of the concern. Examples- accounts payable, bills payable, bank overdraft and outstanding expenses.

Concepts of Working Capital

There are two concepts of working capital: Gross and Net

- Gross working capital-means the total current assets.
- Net working capital-

1. **Gross Working Capital**: The concept of gross working capital refers to the total value of current assets. In other words, gross working capital is the total amount available for financing of current assets. However, it does not reveal the true financial position of an enterprise. How? A borrowing will increase current assets and, thus, will increase gross working capital but, at the same time, it will increase current liabilities also. As a result, the net working capital will remain the same. This concept is usually supported by the business community as it raises their assets (current) and is in their advantage to borrow the funds from external sources such as banks and the financial institutions. In this sense, the working capital is a financial concept. As per this concept:

Gross Working Capital = Total Current Assets

2. **Net Working Capital**: The net working capital is an accounting concept which represents the excess of current assets over current liabilities. Current assets consist of items such as cash, bank balance, stock, debtors, bills receivables, etc. and current liabilities include items such as bills payables, creditors, etc. Excess of current assets over current liabilities, thus, indicates the liquid position of an enterprise. The ratio of 2:1 between current assets and current liabilities is considered as optimum or sound. What this ratio implies is that the firm/ enterprise have sufficient liquidity to meet operating expenses and current liabilities. It is important to mention that net working capital will not increase with every increase in gross working capital. Importantly, net working capital will increase only when there is increase in current assets without corresponding increase in current liabilities.

Types of Working Capital

On the basis of periodicity:

The requirements of working capital are continuous. More working capital is

required in a particular season or the peck period of business activity. On the basis of

periodicity working capital can be divided under two categories as under:

a) Permanent working capital

b) Variable working capital

(a) Permanent working capital: This type of working capital is known as Fixed Working Capital. Permanent working capital means the part of working capital which is permanently locked up in the current assets to carry out the business smoothly.

The minimum amount of current assets which is required to conduct the business smoothly during the year is called permanent working capital. For example, investments required to maintain the minimum stock of raw materials or to cash balance. The amount of permanent working capital depends upon the size and growth of company. Fixed working capital can further be divided into two categories as under:

(I) Regular Working capital:
Minimum amount of working capital required to keep the primary circulation. Some amount of cash is necessary for the payment of wages, salaries etc.
(II) Reserve Margin Working capital:
Additional working capital may also be required for contingencies that may arise any time. The reserve working capital is the excess of capital over the needs of the regular working capital is kept aside as reserve for contingencies, such as strike, business depression etc.
(b) Variable or Temporary Working Capital:
The term variable working capital refers that the level of working capital is temporary and fluctuating. Variable working capital may change from one assets to another and changes with the increase or decrease in the volume of business. The variable working capital may also be subdivided into following two sub-groups.
1. Seasonal Variable Working capital:
Seasonal working capital is the additional amount which is required during the active business seasons of the year. Raw materials like raw-cotton or jute or sugarcane are purchased in particular season. The industry has to borrow funds for short period. It is particularly suited to a business of a seasonal nature. In short, seasonal working capital is required to meet the seasonal liquidity of the business.
2. Special variable working capital:
Additional working capital may also be needed to provide additional current assets to meet the unexpected events or special operations such as extensive marketing campaigns or carrying of special job etc
2) **On the basis of concept**: on the basis of concept working capital is divided into two categories as under:
(A)Gross Working Capital:
Gross working capital refers to total investment in current assets. The current assets employed in business give the idea about the utilization of working capital and idea about the economic position of the company. Thus, gross working capital the amount of funds invested in different current assets. Gross working capital concepts is popular and acceptable concept in the field of finance.
(B) Net Working Capital:
Net working capital means current assets minus current liabilities. The difference between current assets and current liabilities is called the net working capital. If the net working capital is positive business is able to meet its current liabilities. Net working capital concept provides the measurement for determining the creditworthiness of company.

Components of Working capital

a. Inventory Management

Inventories basically include the raw materials, WIP products, and finished goods manufactured or bought from the suppliers. Where the company buys excessive stock, it places a heavy burden on the finances. Similarly, if the inventory isn't available on time, it will lead to the loss of sales. Therefore, inventory management involves the control of the stock that is manufactured/ bought for sale in the normal course of business.

b. Accounts Receivable

Accounts Receivable refers to the amount that is due when sales take place and is allowed by the buyers to the seller while conducting business on credit. It is an anticipation of the payment that has to be received by the debtors within the time-period allowed for making the payment. The process of account receivable management requires effective and accurate handling to ensure the collection of accounts receivable in time. As accounts receivables form a major part of the company's asset, it leads to the generation of cash in-flow for the business.

c. Accounts Payable

Account Payable can aptly be called a short-term debt payable within short periods of time and is treated as a current liability by the entity that has to make the payment. It is the money owed by the businesses to its suppliers for the goods purchased or services consumed. The accounts payable needs to be managed effectively as it enhances the short term cash flow position of the business. This can be done by deciding on an optimal timing for payments to the supplier.

d. Cash Management

Cash and Cash equivalents is the most important current assets that be managed under working capital management. Being the most liquid form of asset, it has to be effectively and efficiently managed to maintain company's financial stability, meet the unexpected expenses and also handle the regular operational expenses on time (such as payroll). Further managing cash rightly also helps to maintain company's' rating and avoid insolvency.Here cash includes, both cash and cash equivalents.

FACTORS DETERMINING THE COMPOSITION OF WORKING CAPITAL

The following factor determine the amount of working capital

1. Nature of Companies: The composition of an asset is a function of the size of a business and the companies to which it belongs. Small companies have smaller proportions of cash, receivables and inventory than large corporation. This difference becomes more marked in large corporations. A public utility, for example, mostly employs fixed assets in its operations, while a merchandising department depends generally on inventory and receivable. Needs for working capital are thus determined by the nature of an enterprise.

2. Demand of Creditors: Creditors are interested in the security of loans. They want their obligations to be sufficiently covered. They want the amount of security in assets which are greater than the liability.

3. Cash Requirements: Cash is one of the current assets which are essential for the successful operations of the production cycle. Cash should not adequate and properly utilized. It would be wasteful to hold excessive cash. A minimum level of cash is always required to keep the operations going. Adequate cash is also required to maintain good credit relation. Richards Osbom has pointed out that cash has a universal liquidity and acceptability. Unlike illiquid assets, its value is clear cut and defines.

4. Nature and Size of Business: The working capital requirements of a firm are basically influenced by the nature of its business. Trading and financial firms have a very less investment in fixed assets, but require a large sum of money to be invested in working capital. Retail stores, for example, must carry large stocks of a variety of goods to satisfy the varied and continues demand of their customers. Some manufacturing business, such as tobacco manufacturing and construction firms also have to invest substantially in working capital and a nominal amount in the fixed assets. In contrast, public utilities have a very limited need for working capital and have to invest abundantly in fixed assets. Their working capital requirements and nominal because they have cash sales only and supply services, not product. Thus, no funds will be tied up in debtors and inventories. The working capital needs of most of the manufacturing concerns fall between the two extreme requirements of trading firms and public utilities. Such concerns have to make adequate investment in current assets depending upon the total assets structure and other variables. The size of business also has an important impact on its working capital needs. Size may me measured in terms of the scale of operation. A firm with larger scale of operation will need more working capital than a small firm.

5. Time: The level of working capital depends upon the time required to manufacturing goods. If the time is longer, the size of working capital is great. Moreover, the amount of working capital depends upon inventory turnover and the unit cost of the goods that are sold. The greater this cost, the bigger is the amount of working capital.

6. Volume of Sales: This is the most important factor affecting the size and components of working capital. A firm maintains current assets because they are needed to support the operational activities which result in sales. They volume of sales and the size of the working capital are directly related to each other. As the volume of sales increase in the investment of working capital-in the cost of operations, in inventories and receivables.

7. Terms of Purchases and Sales: If the credit terms of purchases are more favorable and those of sales liberal, less cash will be invested in inventory. With more favorable credit terms, working capital requirements can be reduced. A firm gets more time for payment to creditors or suppliers. A firm which enjoys greater credit with banks needs less working capital.

8. Inventory Turnover: If the inventory turnover is high, the working capital requirements will be low. With better inventory control, a firm is able to reduce its working capital requirements. While attempting this, it should determine the minimum level of stock which it will have to maintain throughout the period of its operations.

9. Receivable Turnover: It is necessary to have an effective control of receivables. A prompt collection of receivables and good facilities for setting payable results into low working capital requirements.

10. Business Cycle: Business expands during periods of prosperity and declines during the period of depression. Consequently, more working capital required during periods of prosperity and less during the periods of depression. During marked upswings of activity, there is usually a need for larger amounts of capital to cover the leg between collection and increased sales and to finance purchases of additional materials to support growing business activity. Moreover, during the recovery and prosperity phase of the business cycle, prices of raw materials and wages tend to rise and require additional funds to carry even the same physical volume of business. In the downswing of the cycle, there may be a brief period when collection difficulties and declining sales together cause embarrassment by the resulting failure to replenish cash. Later, as the depression runs its course, the concern may find that it has a larger amount of working capital on hand than current business volume may justify.

11. Value of Current Assets: Decreases in the real value of current assets as compared to their book value reduced the size of the working capital. If the real value of current assets increases, there is an increase in working capital.

12. Variations in Sales: A seasonal business requires the maximum amount of working capital for a relatively short period of time.

13. Production Cycle: The time taken to convert raw materials into finished products is referred to as the production cycle or operating cycle. The longer the production cycle, the greater is the requirements of the working capital. An utmost care should be taken to shorten the period of the production cycle in order to minimize working capital requirements.

14. Credit Control: Credit control includes such factors as the volume of credit sales, the terms of credit sales, the collection policy, etc. with a sound credit control policy, it is possible for a firm to improve in cash inflow.

15. Liquidity and Profitability: If a firm desire to take a greater risk for bigger gains or losses, it reduces the size of its working capital in relation to its sales. If it is interested in improving its liquidity, it increase the level of its working capital. However, this policy is likely to result in a reduction of the sales volume, and therefore, of profitability. A firm, therefore, should choose between liquidity and profitability and decide about its working capital requirements accordingly.

16. Inflation: As a result of inflation, size of the working capital is increased in order to make it easier for a firm to achieve a better cash inflow. To some extent, this factor may be compensated by the rise in selling price during inflation.

17. Seasonal Fluctuations: Seasonal fluctuations in sales affect the level of variable working capital. Often, the demand for products may be of a seasonal nature. Yet inventories have got to be purchased during certain seasons only. The size of the working capital in one period may, therefore, be bigger than that in another.

18. Profit Planning and Control: The level of working capital is decided by the management in accordance with its policy of profit planning and control. Adequate profit assists in the generation of cash. It makes it possible for the management to plough back a part of its earnings in the business and substantially build up internal financial resources. A firm has to

plan for taxation payments, which are an important part of working capital management. Often the dividend policy of a corporation may depend upon the amount of cash available to it.

19. Repayment Ability: A firm's repayment ability determines level of its working capital. The usual practices of a firm are to prepare cash flow projections according to its plans of repayment and to fix working capital levels accordingly.

20. Cash Reserves: It would be necessary for a firm to maintain some cash reserve to enable it to meet contingent disbursements. This would provide a buffer against abrupt shortages in cash flows.

21. Operational and Financial Efficiency: Working capital turnover is improved with a better operational and financial efficiency of a firm. With a greater working capital turnover, it may be able to reduce its working capital requirements.

22. Change in Technology: Technological developments related to the production process have a sharp impact on the need for working capital.

23. Firm's Policies: These affect the level of permanent and variable working capital. Changing in credit policy, production policy, etc are bound to affect the size of working capital.

24. Activities of the Firms: A firm's stocking on heavy inventory or selling on easy credit terms calls for a higher level of working capital for it than for selling services or making Cash sales.

LIQUIDITY VERSUS PROFITABILITY: RISK-RETURN

The firm would make just enough investment in current assets, if it were possible to estimate working capital needs exactly. Under perfect certainty, the current assets holdings would be at the minimum level. A ledger investment in current assets under certainty would mean a low rate or return investment for the firm, as excess investment in current assets will not earn enough return. A smaller investment in current assets, on the other hand, would mean interrupted production and sales, because of frequent stock-outs and inability to creditor in time to restrictive credit policy.

As it is not possible to estimate working needs accurately, the firm must decide about the levels of current assets to be carried. The current assets holdings of the firm will depend upon its working capital policy. It may follow a conservative or an aggressive policy. These polices have different risk-return implications. A conservative policy means lower return and risk, while an aggressive policy produces higher return and risk.

The two important aims of the working capital management are: profitability and solvency. Solvency, used in the technical sense, refers to the firm's continuous ability to meet maturing obligations. Lenders and creditors expected prompt settlement of their claims as and when due. To ensure solvency, the firm maintains a relatively large investment in current assets holdings. If the firm maintains a relatively large investment in current assets, it will have no difficulty in paying the claims of the creditors when they become due and will be able to fill all sales orders and ensure smooth production. Thus, a liquid firm has less risk of insolvency; that is, it will hardly experiences a cash shortage or stock-outs. However, there is accost associated

with maintaining a sound liquidity position. A considerable amount of the firm's funds will be tied up in current assets. And to the extent this investment is idle, the firm's profitability will suffer. To have high profitability, the firm may sacrifice solvency and maintain a relatively low level of current assets. When the firm does so, its profitability will improve as less funds are tied up in idle current assets, but its solvency would be threatened and would be exposed to greater risk of cash shortage and stock-outs.

APPROACHES/THEORIES TO WORKING CAPITAL MANAGEMENT

The following three important approaches are applied in practice.

A.) **Matching Approach:**

The firm can adopt a financial plan which involves the matching of the expected life of assets with the expected life of the source of funds raised to finance assets. Thus, a ten years loan may be raised to finance a plant with an expected life of ten years; stock to be sold thirty day may be financed with a thirty-day bank loan or so on. Thus, when the firm follows matching approach, long-term financing will be used to finance fixed assets and permanent current assets. As the firm's fixed assets and permanent current asset levels increase, the long term-term financing level also increases. When temporary current assets level increase, short-term negotiated financing increase, and when the firm has no temporary current assets, it is also has no short-term negotiated financing.

B) **Conservative Approach:**

Conservative financing includes those plans that use more long-term financing than is needed under a matching approach. The firm is financing a portion of its temporary current assets requirements with long-term financing. Also, in periods when the firm has no temporary current assets, the firm has excess financing available that will be invested in marketable securities. These plans are called conservative because they involve relatively heavy use of long-term financing.

C) **Aggressive Approach**: A firm may be aggressive in financing its assets. An aggressive policy is said to be followed by the firm when it uses more short-term financing than warranted by the matching plan. Under an aggressive policy, the firm finances a part of its permanent current assets with short-term financing. Some extremely aggressive firms may even finance a part of their fixed assets with short-term financing. The relatively more use of short-term financing makes the firm more risky.

THREE TYPES OF WORKING CAPITAL POLICIES

Based on the attitude of the finance manager towards risk, profitability and liquidity, the working capital policies can be divided into following three types.

RESTRICTED POLICY

In restricted policy, the estimation of current assets for achieving targeted revenue is done very aggressively without considering for any contingencies and provisions for any unforeseen event. After deciding, these policies are forcefully implemented in the organization without tolerating any deviations. In the diagram, point R represents the restricted policy which attains the same level of revenues with lowest current assets. Adopting this policy would result in an advantage of the lower working capital requirement due to the lower level of current assets. This saves the interest cost to the company and which in turn produces higher profitability i.e. higher return on investment (ROI). On the other hand, there is the

disadvantage in the form of high risk due to very aggressive policy. That is why; it is also called as aggressive working capital policy.

RELAXED POLICY

Relaxed policy is just the opposite of restricted policy. In this policy, the estimation of current assets for achieving the targeted revenue is prepared after careful consideration of uncertain events such as seasonal fluctuations, a sudden change in the level of activities or sales etc. After the reasonable estimates also, a cushion to avoid any unforeseen circumstances is left to avoid the maximum possible risk. In the diagram, it represents the point Rx which uses the highest level of current assets for achieving the same level of sales.

The companies having relaxed working capital policies assume an advantage of almost no risk or low risk. This policy guarantees the entrepreneur of the smooth functioning of the operating cycle. We know that earnings are more important than higher earnings. On the other hand, there is a disadvantage of lower return on investment because higher investment in the current assets attracts higher interest cost which in turn reduces profitability. Because of its conservative nature, this policy is also called as conservative working capital policy.

MODERATE POLICY

Moderate policy is a balance between the two policies i.e. restricted and relaxed. It assumes characteristics of the both the policies. To strike a balance, moderate policy assumes risk which is lower than restricted and higher than conservative. In profitability front also, it lies between the two. The biggest benefit of this policy is that it has reasonable assurance of smooth operation of working operating capital cycle with moderate profitability.

Working capital policies can be further framed for each component of net working capital i.e. cash, accounts receivable, inventory and accounts payable. Cash policies can be to maintain an appropriate level of cash. When the level is high, it should be invested in liquid investments for short term and vice versa. Accounts receivable policy may state about payment terms, credit period, credit limit, etc. Inventory policy may speak of minimizing the levels of inventory till the point it poses any risk to the satisfaction of customer demands. Accounts payable policies include policies of payment terms, quality terms, return policies, etc.

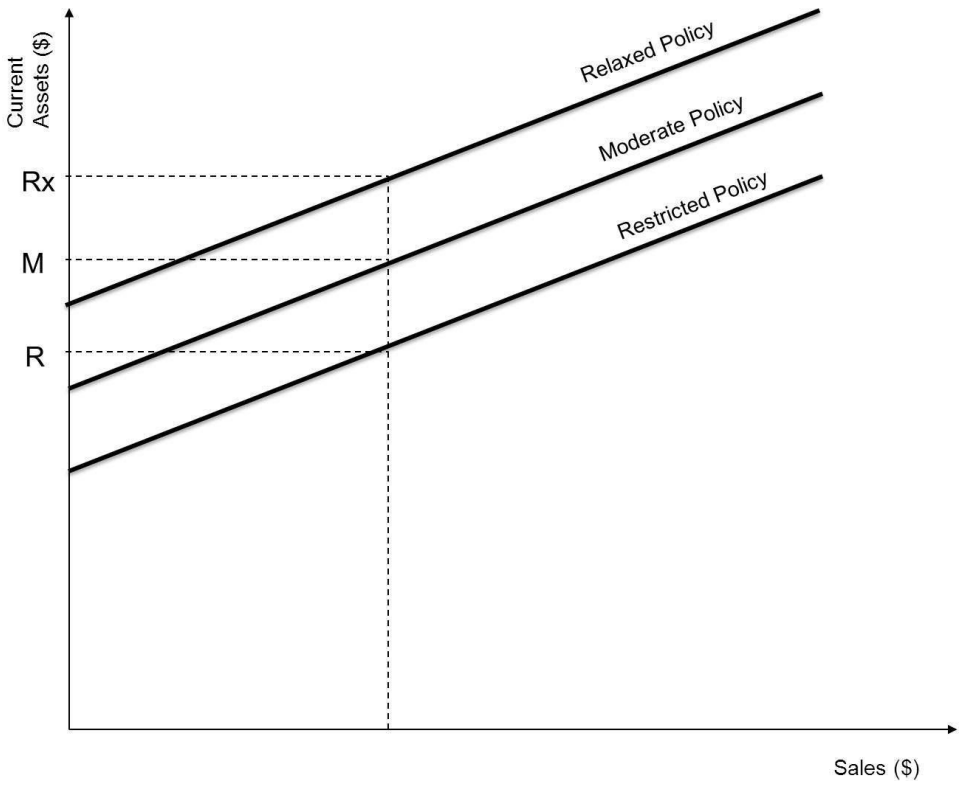


FIG-1 Policies of Working capital Management

TECHNIQUES TO ANALYSE WORKING CAPITAL

There are several methods to conduct a working capital analysis, these include:

A. Ratio Analysis

This is a simple arithmetic view of the relationship between numbers. It is used to measure the short-term liquidity of the firm.

Liquidity Ratio

Ratio	Formula	Description
Current ratio	Current Assets/ Current Liabilities	Also known as the Working Capital Ratio and measures the short-term financial health of a company.
Acid Test Ratio/ Quick Ratio	Liquid Assets/Current Liabilities	Measures if an asset can be liquidated to cash in a short period of time without the loss of value.

Cash Position Ratio/ Absolute Liquid Ratio	[(cash & Bank) + short-term securities]/Current Liabilities	Includes cash in hand and that in the bank and the temporary investments including marketable securities. This ratio must ideally be 50 percent.

Inventory Turnover Ratio = Cost of Goods Sold/Average Inventory at Cost

When the cost of goods sold is not known one may look at the following numbers:

Ratio	Formula	Description
Inventory Turnover Ratio	Cost of Goods Sold/Average Inventory at Cost	When the cost of goods sold is not known one may look at the other formulas
	Net Sales/Average Inventory at Cost	
	Cost of Goods Sold / Average Inventory at Selling Price	

FORECASTING / ESTIMATION OF WORKING CAPITAL REQUIREMENTS

Factors to be considered

- **Total costs incurred on *materials, wages and overheads***
- **The *length of time* for which raw materials remain in stores before they are issued to production.**
- **The length of the production cycle or WIP, i.e., *the time taken for conversion of RM into FG.***
- **The *length of the Sales Cycle* during which FG are to be kept waiting for sales.**
- **The average period of *credit allowed to customers.***
- **The amount of cash required to pay day-to-day expenses of the business.**
- **The amount of cash required for advance payments if any.**
- **The average period of credit to be allowed by suppliers.**
- **Time – lag in the payment of wages and other overheads.**

PROFORMA - WORKING CAPTIAL ESTIMATES

Statement of Working Capital Requirements

		Rs.	Rs.
	Current Assets (Estimated values)		
	Trade Investment	×××	
	Stock-in-trade	×××	
	Sundry Debtors	×××	
	Bills Receivable	×××	
	Cash at Bank	×××	
	Cash in hand	×××	
	Prepaid expenses	×××	
	Total Current Assets [A]		×××
Less :	**Current Liabilities** (Estimated values)		
	Sundry creditors	×××	
	Bank overdraft (unsecured)	×××	
	Bills payable	×××	
	Provision for Taxation	×××	
	Outstanding expenses	×××	
	Total Current liabilities [B]		×××
	Working capital required [A – B]		×××

Proforma Statement of Working Capital Requirement Forecast

Particulars	Period in months or weeks	Total	Current Assets				Current Liabilities : Creditors and others
			Raw materials	Work-in-progress	Finished goods	Debtors	
		Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
1. Materials :							
in store/stock	×		×				
In work-in-progress	×			×			
In finished goods	×				×		
Credit to debtors	×					×	
	×						
Less : Credit from supplier or creditors	×						×
Net block	×	×					
2. Wages/Labour :							
In work-in-progress	×			×			
In finished goods	×				×		
Credit to debtors	×					×	
	×						
Less : Lag in payment of wages	×						
Net block	×	×					×
3. Overheads :							
In work-in-progress	×			×			
In finished goods	×				×		
Credit to debtors	×					×	
	×						×
Less : Lag in payment of overheads	×						
Net block	×	×					
4. Profit :							
Credit to debtors	×						
	×	×				×	
		×	×	×	×	×	×

ESTMATION OF WORKING CAPITAL BY OPERATING CYCLE METHOD

The working capital requirements of a firm depends, to a great extent upon the operating cycle of the firm. The operating cycle may be defined as the time duration starting from the procurement of goods or raw materials and ending with the sales realization.

The length and nature of the operating cycle may differ from one firm to another depending upon the size and nature of the firm.

The operating cycle of a firm consists of the time required for the completion of the chronological sequence of some or all of the following-

- Procurement of raw materials and services.
- Conversion of raw materials into work-in-progress.
- Conversion of work-in-progress into finished goods.
- Sale of finished goods.
- Conversion of receivables into cash.

The formula to calculate operating cycle

Operating cycle = Age of inventory + collection period

Net operating cycle = Age of inventory + collection period – deferred payments

For calculating net operating cycle, various conversion periods may be calculated as follows:

Raw material cycle period (RMCP)= (Average Raw material stock/Total raw material Consumable) x 365

Working progress cycle period (WPCP) = (Average work in progress/ Total cost of Production) x 365

Finished goods cycle period (FGCP) = (Average finished goods/Total cost of goods Sold) x 365

Accounts receivable cycle period (ARCP) = (Average Account receivable/Total of sales) x 365

Accounts payable cycle period (APCP) = (Average account payable/ Total credit purchase) x 365

where,

Total credit purchase = cost of goods sold + ending inventory – beginning of inventory

\For above calculations, the following points are essential

1. The average value is the average of opening balance and closing balance of the respective items. In case the opening balance is not available, only the closing balance is taken as the average.

2. The figure 365 represents number of days in a year. Sometimes even 360 days are considered.

3. The calculation of RMCP, WPCP and FGCP the denomination is taken as the total cost raw material consumable, total cost of production total, cost of goods sold respectively since they form respective end products.

On the basis of the above, the operating cycle period:

Total operating cycle period (TOCP) = RMCP + WPCP + FGCP + ARCP

Net operating cycle period(NOCP) = TOCP-DP(deferred payment) (APCP)

The operating cycle for individual components are not constant in the growth of the business. They keep on changing from time to time, particularly the Receivable Cycle Period and the Deferred Payment. But the company tries to retain the Net Operating Cycle Period as constant or even less by applying some requirements such as inventory control and latest technology in production. Therefore regular attention on the firm's operating cycle for a period with the previous period and with that of the industrial average cycle period may help in maintaining and controlling the length of the operating cycle.

Working Capital Management

FINANCIAL MANAGEMENT-UNIT 4

Nature of Working Capital

- Working capital management is concerned with the problems that arise in attempting to manage the current assets, the current liabilities and the interrelations that exist between them.
- Current assets refer to those assets which in the ordinary course of business can be, or will be, converted into cash within one year without undergoing a diminution in value and without disrupting the operations of thefirm.

  Examples- cash, marketable securities, accounts receivableand inventory.
- Current liabilities are those liabilities which are intended, at their inception, to be paid in the ordinary course of business, within a year, out of the current assets or the earnings of the concern.

  Examples- accounts payable, bills payable, bank overdraftand outstanding expenses.

Objective of Working Capital Management

- The goal of working capital management is to manage the firm's current assets and liabilities in such a way that a satisfactory level of working capital is maintained.
- The interaction between current assets and current liabilities is, therefore the main theme of the theory of the working capital management.

Concepts and Definitions of Working Capital

There are two concepts of working capital:

Gross and Net.

- Gross working capital- means the total current assets.
- Net working capital- can be defined in two ways-
  - The difference between current assets and current liabilities.
  - The portion of current assets which is financed with long term funds.

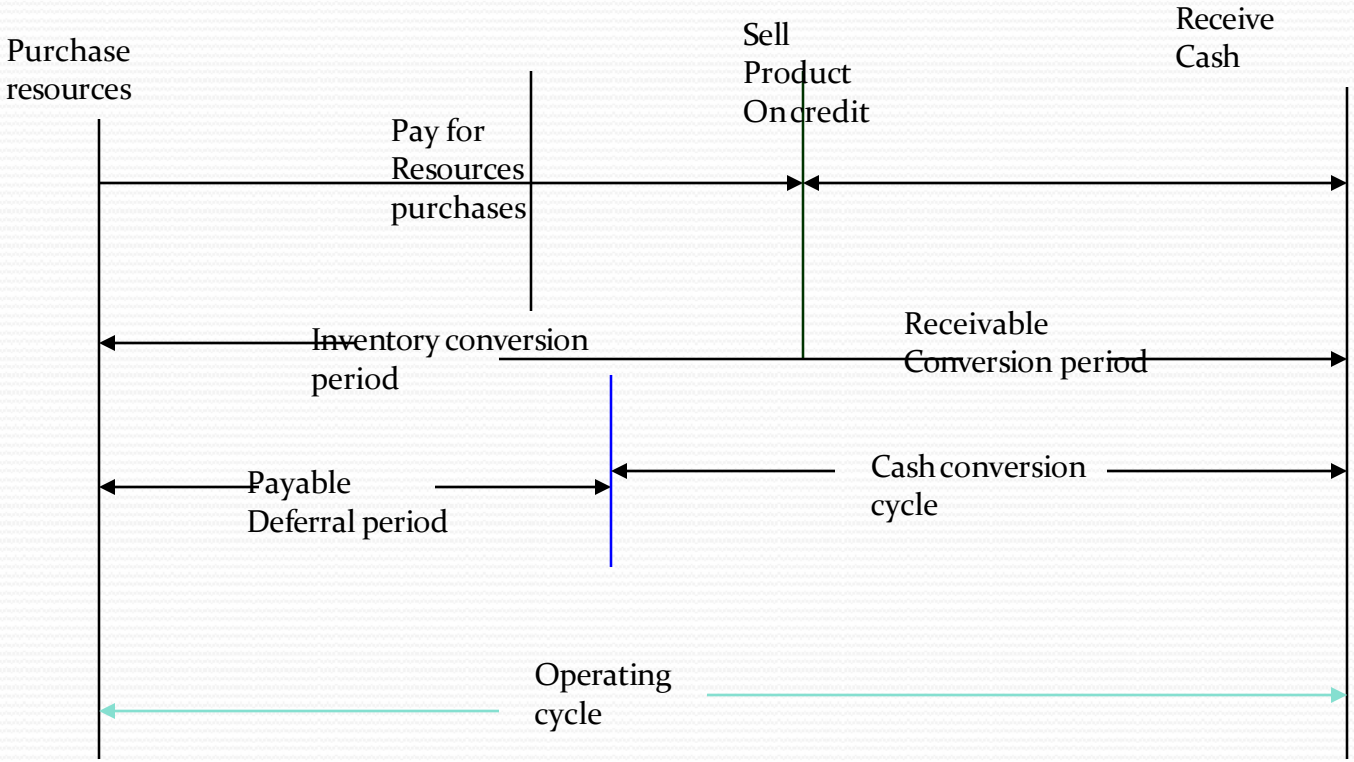
Determining Financing-mix

- There are two sources from which funds can be raised for current assets financing-
  - Short term sources, like current liabilities and,
  - long term sources, such as share capital, long term borrowings, internally generated resources like retained earnings, etc.

The Operating-cycle and Working Capital Needs

- The working capital requirements of a firm depends, to a great extent upon the operating cycle of the firm. The operating cycle may be defined as the time duration starting from the procurement of goods or raw materials and ending with the sales realization.
- The length and nature of the operating cycle may differ from one firm to another depending upon the size and nature of the firm.
- The operating cycle of a firm consists of the time required for the completion of the chronological sequence of some or all of the following-
  - Procurement of raw materials andservices.
  - Conversion of raw materials intowork-in-progress.
  - Conversion of work-in-progress into finishedgoods.
  - Sale of finished goods.
  - Conversion of receivables intocash.

Operating cycle of a typical company



- **Inventory conversionperiod**

$$= \frac{\text{Avg. inventory}}{\text{Cost of sales}/365}$$

- **Receivable conversion period**

$$= \frac{\text{Accounts receivable}}{\text{Annual credit sales}/365}$$

- ***Cash conversion cycle*** = operating cycle – payables deferral period.

- **Payables deferral period**

Accounts payable + Salaries, etc

= _____

__(Cost of sales + and admn. selling, general Expenses)/365

Determinants of Working capital Requirement

- General nature of business
- Production cycle
- Business cyclefluctuations
- Production policy
- Credit policy
- Growth and expansion
- Profit level
- Level of taxes
- Dividend policy
- Depreciation policy
- Price level changes
- Operating efficiency

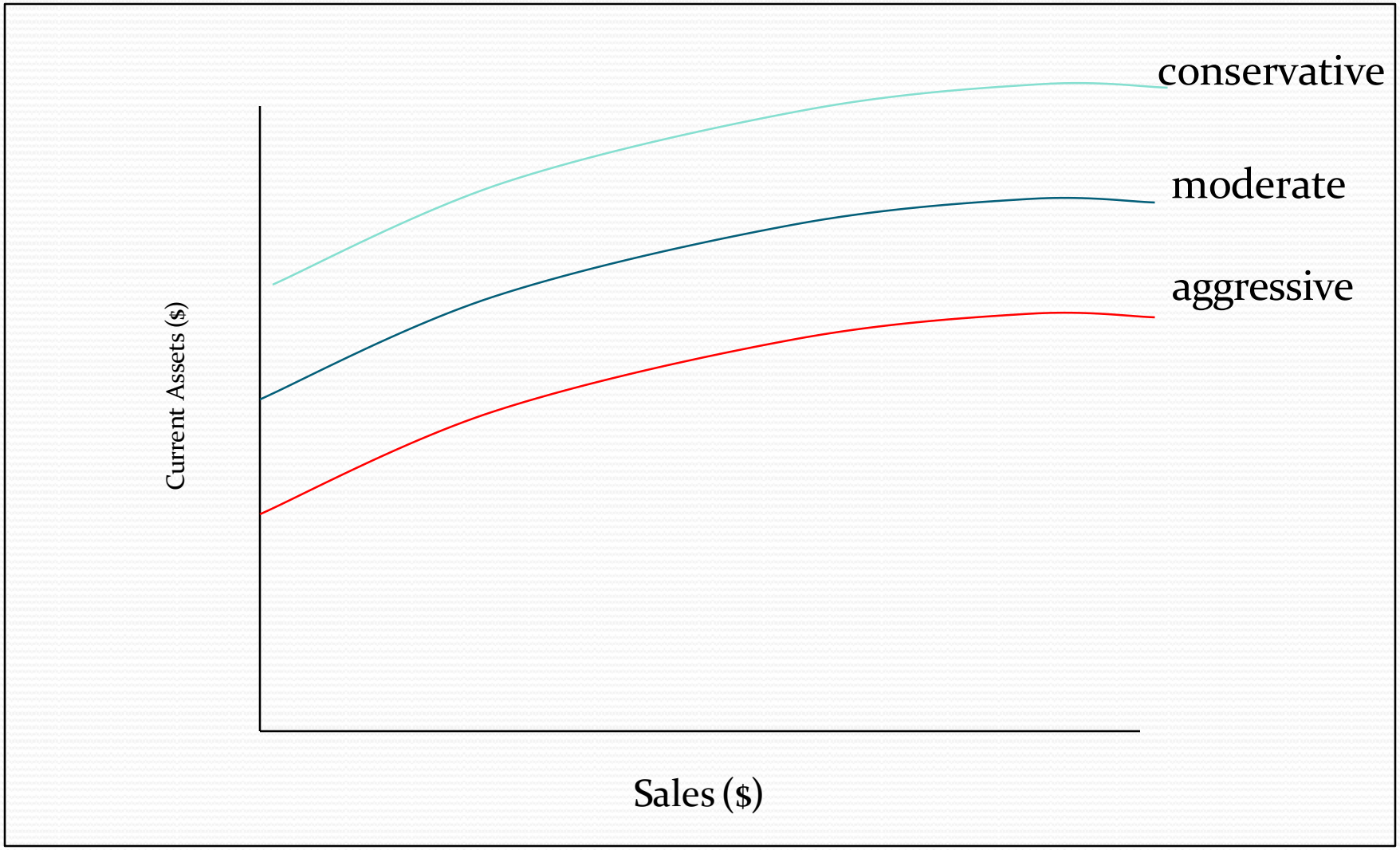
Working capital: Policy and Management

- The working capital management includes and refers to the procedures and policies required to manage the working capital.

  There are three types of working capital policies which a firm may adopti.e.
- Moderate working capital policy
- Conservative working capital policy
- Aggressive working capital policy.

  These policies describe the relationship between the sales level and the level of currentassets.

Three alternative working capital investment policies



Liquidity versus Profitability- A Risk-Return Trade-off

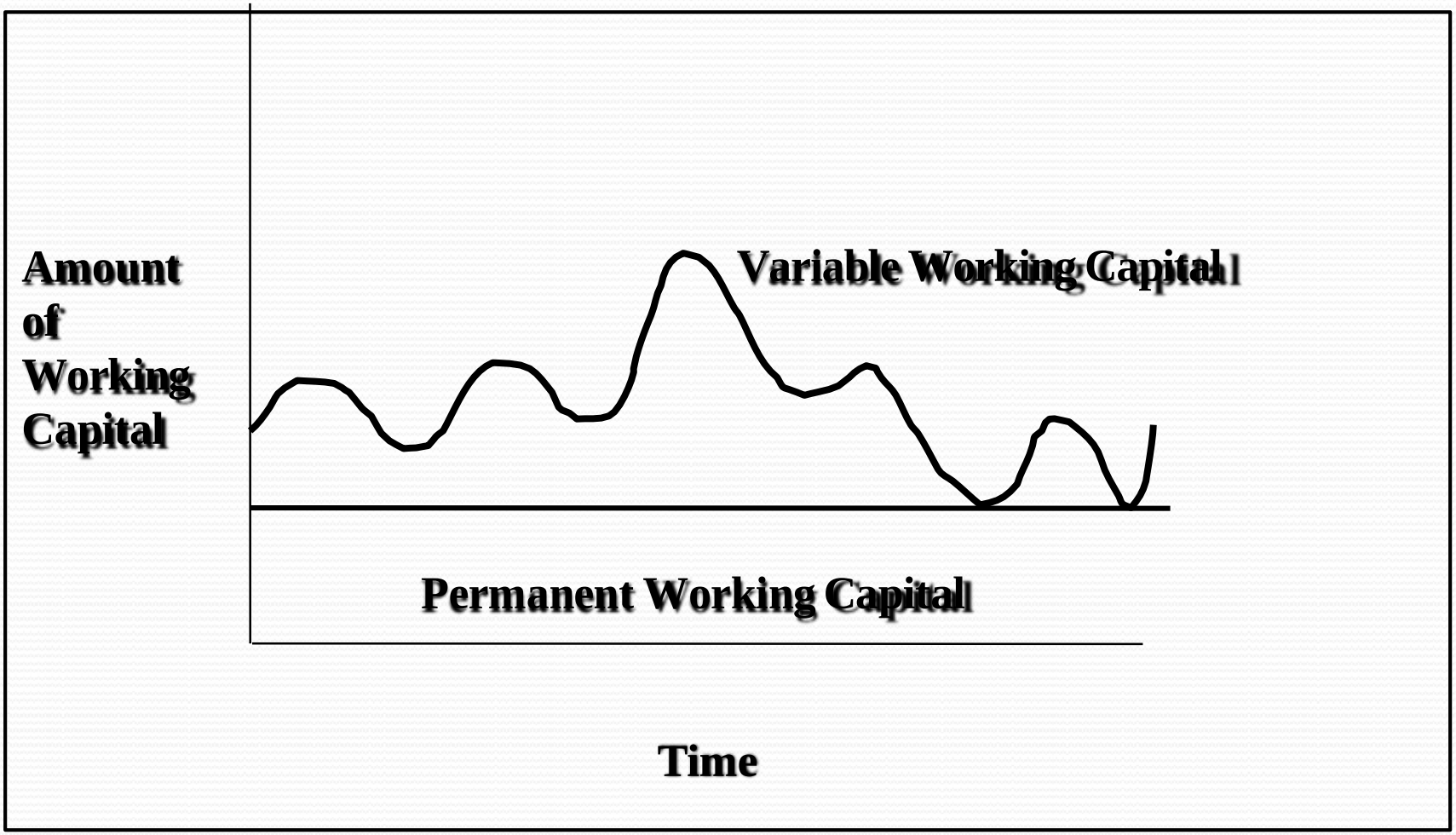
- An important aspect of a working capital policy is to maintain and provide sufficient liquidity to the firm. The decision on how much working capital be maintained involves a trade-off i.e., having a large net working capital may reduce the liquidity-risk facedby the firm, but it can have a negative effect on the cash flows. Therefore, the net effect on the value of the firm should be used to determine the optimal amount of working capital.

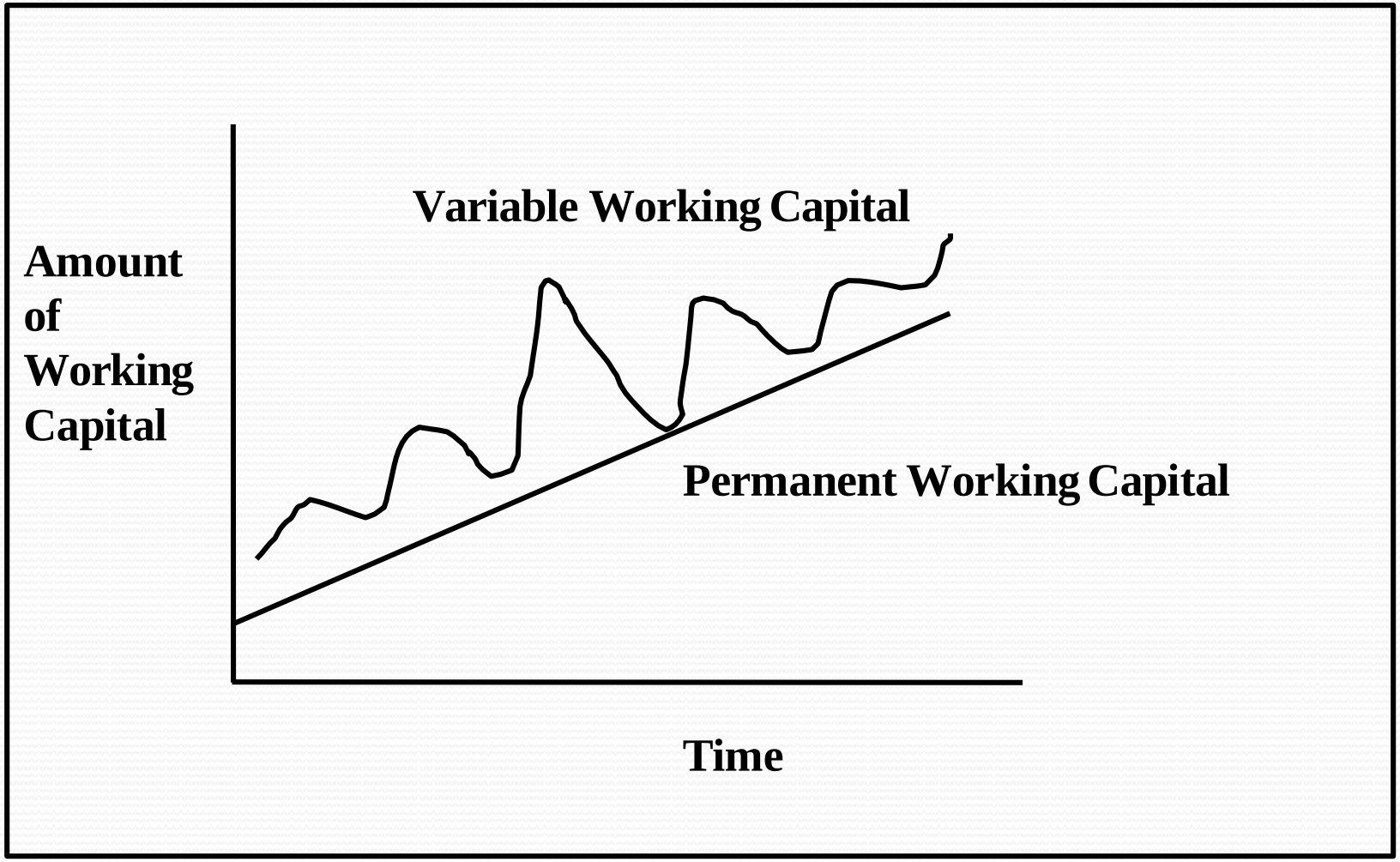
Types of working capital needs

- The working capital need can be bifurcated into permanent working capital and temporary working capital.

- **Permanent working capital-** There is always a minimum level of working capital which is continuously required by a firm in order to maintain its activities like cash, stock and other current assets in order to meet its business requirements irrespective of the level of operations.
- **Temporary working capital-** Over and above the permanent working capital, the firm may also require additional working capital in order to meet the requirements arising out of fluctuations in sales volume. This extra working capital needed to support the increased volume of sales is known as temporary or fluctuating workingcapital.

Difference between permanent & temporary working capital




Variable Working Capital
Amount of Working Capital
Permanent Working Capital
Time

Approaches to determine an appropriate Financing-mix

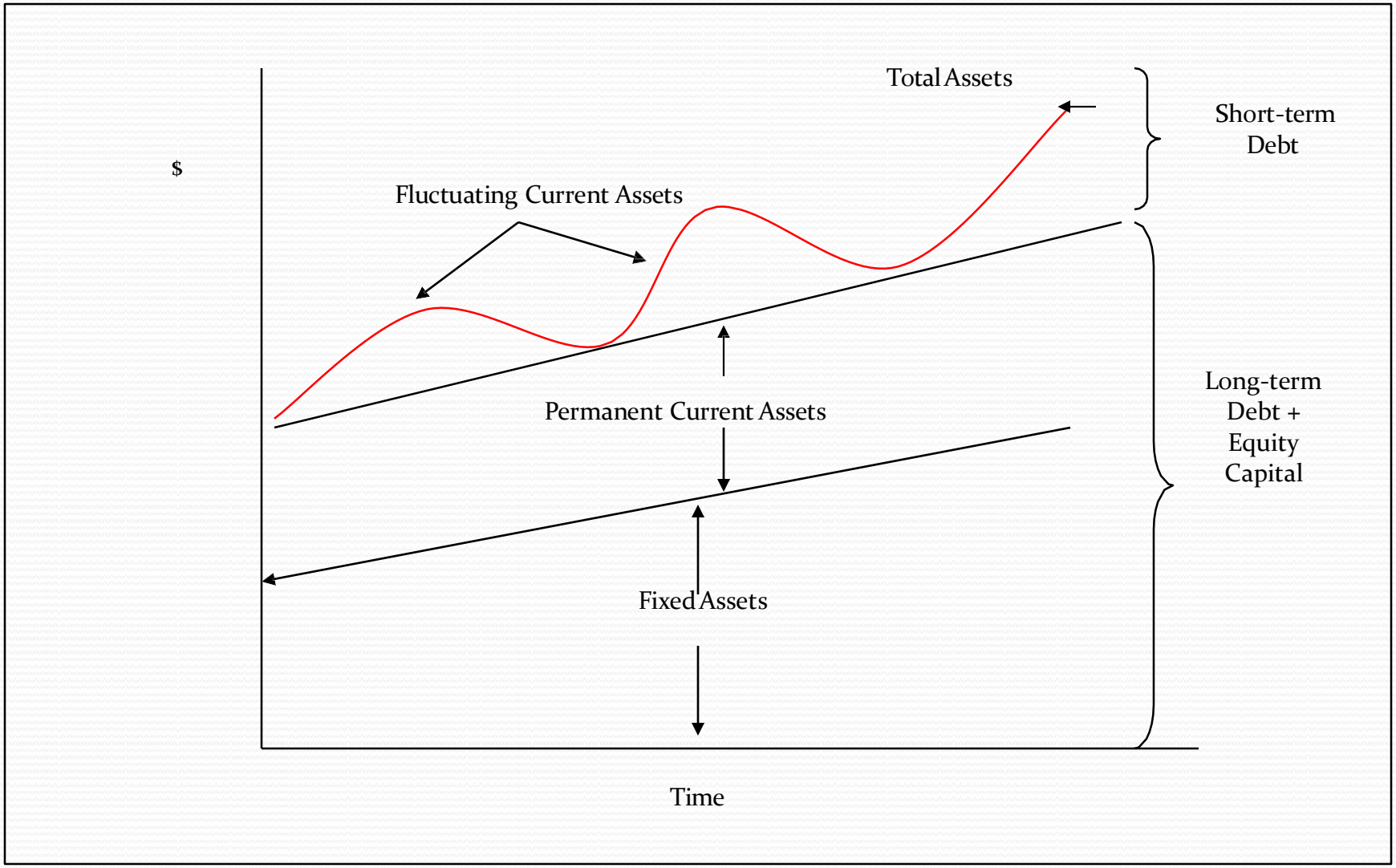
There are three basic approaches to determine an appropriate financing mix:

- Hedging approach, also called the matching approach,
- Conservative approach,
- Aggressive approach.

Hedging Approach/ Matching Approach

- According to this approach, the maturity of the sources of the funds should match the nature of the assets to be financed. For the purpose of analysis, the current assets can be broadly classified into two classes-
  - those which are required in a certain amount for a given level of operation and, hence, do not vary over time.
  - those which fluctuate over time.
- The Hedging approach suggests that long term funds should be used to finance the fixed portion of current assets requirements in a manner similar to the financing of fixed assets.
- The purely temporary requirements, that is, the seasonal variations over and above the permanent financing needs should be appropriately financed with short term funds.
- This approach, therefore, divides the requirements of total funds into permanent and seasonal components, each being financed by a different source.

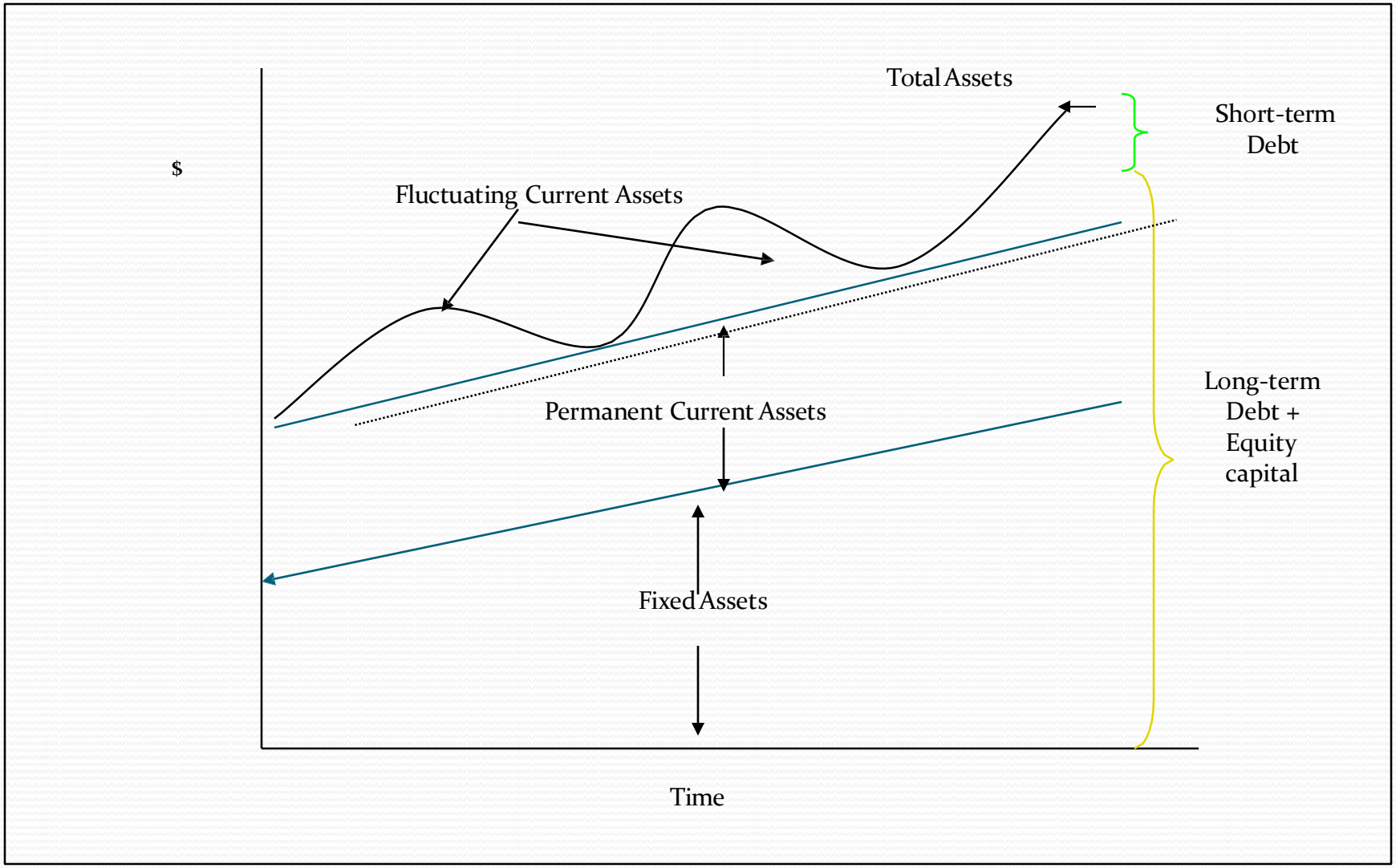
Matching approach to asset financing



Conservative Approach

- This approach suggests that the estimated requirement of total funds should be met from long term sources; the use of short term funds should be restricted to only emergency situations or when there is an unexpected outflow of funds.

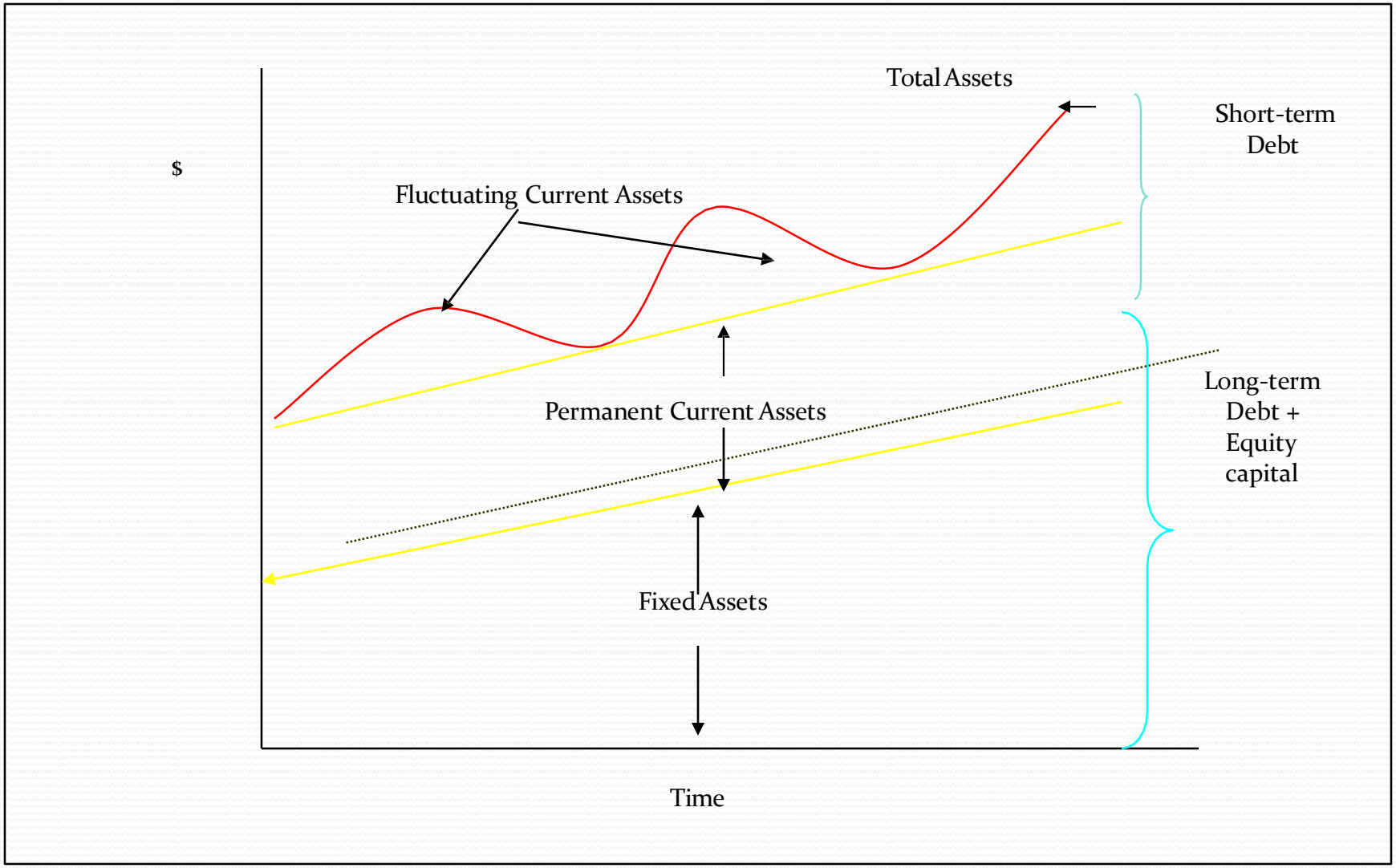
Conservative approach to asset financing



Aggressive approach

- A working capital policy is called an aggressive policy if the firm decides to finance a part of the permanent working capital by short term sources. The aggressive policy seeks to minimize excess liquidity while meeting the short term requirements. The firm may accept even greater risk of insolvency in order to save cost of long term financing and thus in order to earn greater return.
- The trade-off between risk and profitability depends largely on the financial manager's attitude towards risk, yet while doing so he must take care of the following factors-
  - Flexibility of the mix
  - Cost of financing
  - Risk attached with financing mix

Aggressive approach to asset financing



Forecasting / Estimation of Working Capital Requirements

Factors to be considered

- Total costs incurred on *materials, wages and overheads*
- The *length of time* for which raw materials remain in stores before they are issued to production.
- The length of the production cycle or WIP, i.e., *the time taken for conversion of RM into FG.*
- The *length of the Sales Cycle* during which FG are to be kept waiting for sales.
- The average period of *credit allowed to customers.*

- The amount of cash required to pay day-to-day expenses of the business.
- The amount of cash required for advance payments if any.
- The average period of credit to be allowed by suppliers.
- Time – lag in the payment of wages and other overheads.

PROFORMA - WORKING CAPTIAL ESTIMATES

1. TRADING CONCERN

STATEMENT OF WORKING CAPITALREQUIREMENTS

	Amount (Rs.)
Current Assets	
(i) Cash	**----**
(ii) Receivables (For…..Month's Sales)----	**----**
(iii) Stocks (For……Month's Sales)-----	**----**
(iv)Advance Payments if any	**----**
Less : Current Liabilities	
(i) Creditors (For….. Month's Purchases)-	**----**
(ii) Lag in payment of expenses	**-----**
WORKING CAPITAL (CA – CL)	**xxx**
Add : Provision / Margin for Contingencies	**-----**
NET WORKING CAPITALREQUIRED	**XXX**

MANUFACTURING CONCERN

STATEMENT OF WORKING CAPITALREQUIREMENTS

	Amount (Rs.)
Current Assets	
(i) Stock of R M(for ….month's consumption)	-----
(ii)Work-in-progress (for…months)	
(a) Raw Materials	-----
(b) Direct Labour	-----
(c) Overheads	-----
(iii) Stock of Finished Goods (for …month's sales)	
(a) Raw Materials	-----
(b) Direct Labour	-----
(c) Overheads	-----
(iv) Sundry Debtors (for …month's sales)	
(a) Raw Materials	-----
(b) Direct Labour	-----
(c) Overheads	-----
(v) Payments in Advance (if any)	-----
(iv) Balance of Cash for dailyexpenses	-----
(vii)Any other item	-----
Less : Current Liabilities	
(i) Creditors (For….. Month's Purchases)	-----
(ii) Lag in payment of expenses	-----
(iii) Any other	-----
WORKING CAPITAL (CA – CL)xxxx	
Add : Provision / Margin for Contingencies	-----
NET WORKING CAPITALREQUIRED	XXX